

Renormalizability

- Previous L² & Divergences:

 min. Order



- Had to introduce counterterms to obtain something physical
- will the QED counter terms found remove all infinities?
 - yes renormalizable
 - no not renormalizable

Def'n: Renormalizable:

- All UV div. canceled with a finite number of counter terms

- showing QED Manual: Table:

- Note: All UV div. are the ^{if 1 loop} same
whether or not ext- legs are on shell

↳ game counter forces

cancel UV div. of div. graphs
even when in complicated cov. fun.

PI:

- summed over PI graphs in ext. lines
↳ ^{integrated} renormalized propagator

↳ look at PI to figure out UV div.

new equiv. Leth of PI:

- PI if all internal lines have some loop momentum
going through them

graph in computation of

Greenfunction. computed w/ PI graphs w/ off-shell
momenta

∴ if the counter-terms cancel PI div., cancel UV div. in all green
function.

- Degree of Div.:

- $D =$ power of loop momentum

$$\text{Ex: } \int d^4k \times k^{-2} \rightarrow D=2$$

Graph n/D scales as Λ^p & $m \Lambda$ if $D < 0$

21.1: Generalizability of QED:

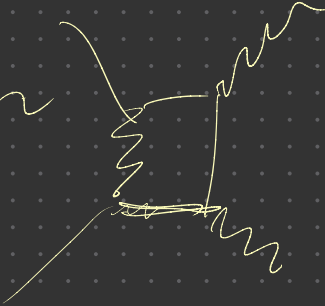
- 1d 2ϕ is point func.
- Non used for Posit

21.1.1: 4-point func

Consider:

$$\langle \bar{\psi} \psi \bar{\psi} \psi \rangle \sim \text{Diagram} \sim \int d^4x \frac{1}{x^2} \frac{1}{x^2} \frac{1}{x^2} \frac{1}{x^2} = \frac{1}{\Lambda^2} = \text{not dim.}$$


- Consider next 1PI:

$$\langle \bar{\psi} \psi A A \rangle \sim \text{Diagram}$$


- Also not UV div.

$\Rightarrow$ not UV div.

- Finally, consider

$$M \sim (AAAA) \sim \begin{array}{c} \text{---} \\ \diagup \quad \diagdown \\ \text{---} \end{array}$$

$$\sim \int d^4x \frac{1}{k} \frac{1}{k} \frac{1}{k} \frac{1}{k} \sim \Lambda^0$$

log-div.

note:

$$M = \epsilon_1^\mu \epsilon_2^\nu \epsilon_3^{\rho\sigma} \epsilon_4^{\alpha\beta} M_{\mu\nu\rho\sigma}$$

(L.T., dim. anal., symm.)

$$\therefore M_{\mu\nu\rho\sigma} = c \ln \Lambda^2 (g_{\mu\nu} g_{\rho\sigma} + g_{\mu\rho} g_{\nu\sigma} + g_{\mu\sigma} g_{\nu\rho}) \Lambda_{\text{m.f.t.}}$$

(c const.)

using Ward identity,

$$0 = q^\mu M_{\mu\nu\rho\sigma} = c \ln \Lambda^2 (q_\nu g_{\rho\sigma} + q_\rho g_{\nu\sigma} + q_\sigma g_{\nu\rho}) + q^\mu \text{ finite}$$

$\Gamma = C=O$

$\Gamma = , \Theta =$, point form

consider new diagrams:

$$(\Gamma^4 A_2 A_0 A_0) =$$



$$\sim \int d^4x \frac{1}{x^4} \frac{1}{x^4} \frac{1}{x^4} \frac{1}{x^4} \frac{1}{x^2} \sim \frac{1}{\Lambda^2}$$

- All of these graphs have $\rightarrow \Gamma$ loop

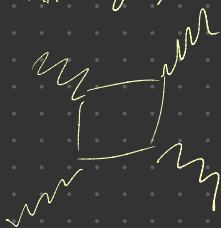
$\rightarrow$ All time $\rightarrow \Gamma$ 0

$\rightarrow$ QED renormalizable for loop

2.1, 3 Higher order loops

- Need to show that adding new loops does not increase log. div.

consider again



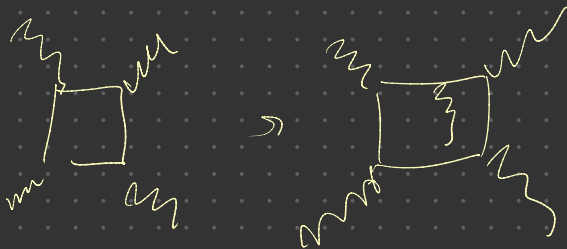
neutral

loops

either

- add photon prop.
- add electron prop.

photon:



then, by 1PI def'n, representative goes through all internal lines

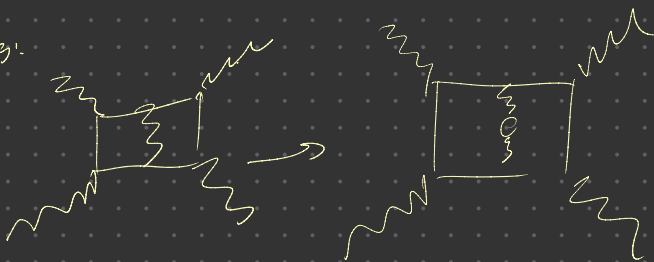
$\therefore$ internal lines get $\frac{1}{h}$ or $\frac{1}{h^2}$

$$\therefore \int d^4x \frac{1}{h} \rightarrow \int d^8x \frac{1}{h^4} \frac{1}{h^2} \frac{1}{h} \frac{1}{h} \sim \Lambda^0$$

- remains the same

- similarly, can add fermion loop w/ 4 photon lines

Eg:



$$\text{Then, } \int d^8x \frac{1}{h^4} \rightarrow \int d^{12}x \frac{1}{h^4} \frac{1}{h^2} \frac{1}{h} \frac{1}{h} \sim \Lambda^0$$

- proof of renormalizability w/renorm induction
- tells two edge case

TPH2. Thm:

- All div. can be removed by counterterms corresponding to superficially div. 1PI amplitudes

This implies

QED is renormalizable

2.2: Non-renormalizable

- Renormalizability desirable for a theory

- caveats =

- very difficult

- cannot predict every observable simply b.c. we can cancel UV div.

- QED breaks at 10^{286} eV

- QCD breaks unknown

or a theory theory

- no UV div. at all

- haven't related any observable to any of these

modern view

limitations of renormalizability is largely irrelevant for physical calculation

Degree of D.V. table:

(A formica & vacuum)

$$D = 4 - \frac{3}{2} P - b$$

$$\langle AA \rangle = \int d^4 u \frac{1}{u} \frac{1}{u} \sim \int^2 D_{0,2} = 2$$

$$\langle \bar{\psi} \psi \rangle = \int d^4 u \frac{1}{u^2} \frac{1}{u} \sim \int^1 D_{2,0} = 1$$

...

21.2.1 D.V. in non renorm. situation: $\dots, E^2)$

- fermions, bosons (+ scalars)
deg. of d.v.:

$$D = 4 - \frac{3}{2} P - b - s$$

may be done w/ log z reduction:

$$\phi^d(\mathbb{R}^d) M \sim \prod_{i=1}^b \int d^4 y_i \frac{1}{y_i^2} \sim \prod_{j=1}^f \int d^4 y_j \frac{1}{y_j^2} \dots$$

$$\times (q_1(y_1) \dots \psi(y_p) A_1(y_1) \dots A_b(y_b))$$

- fermions have b.c.m. $\frac{3}{2}$, photon
green line. dim $\frac{3}{2}$, $\frac{1}{2}$ & $\frac{1}{2}$

same dim. -4

$$\chi[M] = \frac{3}{2}f + 6 - 2b - 2d + 4$$

$$= 4 - \frac{3}{2}f - b$$

suppose strings coupling:

ex:

$$\mathcal{L} \sim \frac{1}{2} \phi (\Box \phi)^2 + g_1 \phi^3 + g_2 \phi^4 \Box \phi^3$$

$\Delta_i \equiv$ mass dim. of coeff. of interaction

$$= [g_i]$$

$$\dim = \Delta_1 = [g_1] = 1, \quad \Delta_2 = [g_2] = -3$$

consider loop w/ n_i insertions of ϕ_{Δ_i}

$$\therefore \int \mu^D g \rightarrow \left(\prod_i g_i^{n_i} \right) \int \mu^{D - \sum_i n_i \Delta_i}$$

degree of $d\mu$

$\therefore$

$$D_{f,b,n_i} = 4 - \frac{3}{2}f - b - \sum_i n_i \Delta_i$$

$\therefore$ if $\Delta_i < 0$, infinite such n_i s.t. $D > 0$

$\therefore$ infinite # of renorms

$\therefore$ non renormalizable

- Non renormalizable can also be from propagators

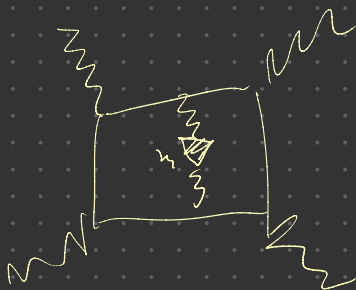
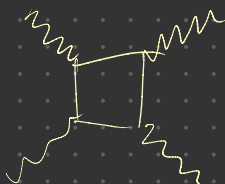
Ex: massive vector boson:

$$i\Pi_{\mu\nu} \sim \frac{-i(g_{\mu\nu} - \frac{p_\mu p_\nu}{p^2})}{p^2 - m^2 + i\epsilon}$$

- At $p^2 \gg m^2$, goes as $\frac{1}{m^2} (?)$

$\therefore$ massive vec. boson has 2 more powers of k^2 than photon

Ex:



$$\int d^4x \frac{1}{k} \frac{1}{k} \frac{1}{k} \frac{1}{k} \sim 1^0 \rightarrow \int d^4x \frac{1}{k} \frac{1}{k} \frac{1}{k} \frac{1}{k} \int d^4x \frac{1}{k^2} \frac{1}{k} \frac{1}{k}$$

$$\sim \frac{1^2}{k^2}$$

Renormalizing the counter terms

- can use original parts of the counter terms

$$\text{Ex: } \lambda \rightarrow \frac{1}{2} \phi (\Box + m^2) \phi + \frac{g}{4!} \phi^2 \Box \phi^2$$

(EFT = 2)

→ 2 point amplitudes

$$\text{X} + \text{X} \sim g p^2 + g^2 (c_1 \Delta^4 + c_2 \Delta^2 + c_3 p^2 m^2)$$

- w/ only counterterm, can only renormalize Δ^2 term

Remove other div

→ larger long range

$$\begin{aligned} \lambda &\rightarrow -\frac{g}{2} \phi (\Box + m^2) \phi + \lambda_p Z_\lambda \phi^4 + g_p Z_g \phi^2 \Box \phi^2 \\ &\quad + k_p Z_k \phi^2 \Box^2 \phi^2 + \dots \end{aligned}$$

Expand

$$Z_\lambda = 1 + \delta_\lambda, \quad Z_g = 1 + \delta_g, \quad Z_k = 1 + \delta_k$$

Counter terms

$$\text{X} \sim \lambda_p \delta_\lambda + g_p \delta_g p^2 + k_p \delta_k p^4$$

Then $\lambda_p \delta_\lambda = -g_p^2 c_1 \Delta^4$, $\delta_g g_p = -g_p^2 c_2 \Delta^2$, $\delta_k k_p = -g_p^2 c_3 m^2$

cancels infinities
creates new but (one repeat process)

Divergent integrals ($h \neq p$)

of form:

$$\int_{\text{dim.}} (p_0 + p_1 \dots p_n) g_1 \dots g_n \frac{d^4 k}{k^2}$$

(in external momenta)

production powers of $\Delta + \log \Delta$

$$\Rightarrow I_{\text{div.}} = \int_{m, n} g_1 \dots g_n (p_0 + p_1 \dots p_n) \left[c_0^{m, n} \Delta + c_1^{m, n} \Delta^2 + c_2^{m, n} \Delta^3 + \dots \right]$$

Wolfe:

Divergences from loop integrals multiply polynomials in ext. momenta

proof (arXiv:1905.0945)

have integral $I(h)$

- suppose at least 1 factor of p in denom.

$\Rightarrow$ diff. not enough: give recurrence in p

$\Rightarrow$ integrals over p producing a polynomial (up to infinitesimal constants)

$$\text{Ex: } I(p) = \int_0^\infty \frac{p \, d^4 k}{k^2 + p}$$

$$I''(p) = \int_0^\infty \frac{2 \, d^4 k}{(k^2 + p)^3} = \frac{1}{p}$$

(integration counts: Δ, c, Δ)

$$\Rightarrow I(p) = p \ln \frac{p}{\Lambda} - p + c_1 \Delta = p \ln p - p (\ln \Delta + 1) + c_1 \Delta$$

- integration constants are same as divergences

- Notes: polynomials in p are what is obtained from tree level terms

$\rightarrow$ All $\Rightarrow$ mel can be made UV finite

21.2.3 Non renormalizable theories are predictive

- Tree level terms, higher derivatives have much weaker low energy effect.

Ex:

$$\mathcal{L} \sim -\frac{1}{2} \partial(\Box + m^2)\phi + \lambda \phi^4 + \frac{g_1}{M^2} \partial^2 \phi \partial^2 \phi + \frac{g_2}{M^4} \partial^4 \phi^2$$

($M \gg 1$, g_i dimensionless)

- consider unrenormalizable $\mathcal{L} \sim \mathcal{L}^{(4)}$ as Oms. of energy scale E

$\Rightarrow$ polynomial in E

$\Rightarrow$ Expand around $E=0$: $\mathcal{L}(\phi) = \chi \phi^2 + g_1 \frac{E^2}{M^2} + g_2 \frac{E^4}{M^4} + \dots$

- At low energy, further terms are less useful (like $\frac{1}{M^6}$)

$\therefore$ Non renormalizable theories are predictive

- Also predictive at loop level

key: UV div. prop to polynomials in p

$\Rightarrow$ terms required to renorm are polynomial in derivatives
 $\Rightarrow$ contact diagrams effects

Ex: effective potential $V(\phi)$ (given by Fourier transform from ϕ^2 part by loop approx)

Ex: $\frac{1}{M^2} \partial^2 \phi$ contribution

$$\mu(p) \sim \text{---} \otimes \text{---} = \frac{1}{p^2} \frac{p^4}{M^2} \frac{1}{p^2} = \frac{1}{M^2}$$

$$\Rightarrow V(\phi) \sim \frac{1}{M^2} g(\phi)$$

- irrelevant at long dist

- more insertions of $\phi \square \phi$, $\phi \square^2 \phi$.

valids

$$V(v) \sim \int d^3 \vec{p} \, e^{i \vec{p} \cdot \vec{x}} \left(\frac{\vec{p}^2}{m^2} \right)^n = \left(\frac{-\Delta}{m^2} \right) \int d^3 \vec{p} \, e^{i \vec{p} \cdot \vec{x}} = \left(\frac{-\Delta}{m^2} \right)^n g^2(\vec{x})$$

- extremely local effects

Loops:

- compute $\ln(\vec{p}^2)$, etc.

$$\rightarrow V(v) = \int d^3 \vec{p} \, e^{i \vec{p} \cdot \vec{x}} \ln \vec{p}^2 = -\frac{1}{4\pi x^2}$$

dimensionless local effects